

Q1.

$$1. \{ (x, y) \in \mathbb{R}_+^2 \mid x/y \leq 1 \} \quad x/y \leq 1 \Leftrightarrow x \leq y$$

$$\text{Pick } P_1 = (x_1, y_1), P_2 = (x_2, y_2)$$

$$P_1, P_2 \text{ both satisfy } x \leq y$$

$$\text{So } P' = \theta P_1 + (1-\theta) P_2 = [\theta x_1 + (1-\theta)x_2, \theta y_1 + (1-\theta)y_2]$$

$$\begin{aligned} \because \begin{cases} x_1 \leq y_1 \\ x_2 \leq y_2 \end{cases} &\Rightarrow \begin{cases} \theta(x_1 - y_1) \leq 0 \\ (1-\theta)(x_2 - y_2) \leq 0 \end{cases} \Rightarrow \theta(x_1 - y_1) + (1-\theta)(x_2 - y_2) \leq 0 \\ &\Leftrightarrow \theta x_1 + (1-\theta)x_2 \leq \theta y_1 + (1-\theta)y_2 \end{aligned}$$

$$\therefore P' \text{ also satisfies } x \leq y$$

This set is convex set

2. similar to Q1, pick P_1, P_2

$$P' = [\theta x_1 + (1-\theta)x_2, \theta y_1 + (1-\theta)y_2]$$

$$\begin{cases} \theta(x_1 - y_1) \geq 0 \\ (1-\theta)(x_2 - y_2) \geq 0 \end{cases} \Rightarrow \theta(x_1 - y_1) + (1-\theta)(x_2 - y_2) \geq 0 \Leftrightarrow \theta x_1 + (1-\theta)x_2 \geq \theta y_1 + (1-\theta)y_2$$

$$\therefore P' \text{ also satisfies } x \geq y \text{ (} x/y \geq 1 \text{)}$$

This set is convex set

$$3. \text{ Pick } P_1, P_2, \quad P' = [\theta x_1 + (1-\theta)x_2, \theta y_1 + (1-\theta)y_2] \quad \{ (x, y) \in \mathbb{R}_+^2 \mid xy \geq 1 \}$$

$$P'_x = \theta x_1 + (1-\theta)x_2, \quad P'_y = \theta y_1 + (1-\theta)y_2$$

$$P'_x P'_y = [\theta x_1 + (1-\theta)x_2] [\theta y_1 + (1-\theta)y_2] = \theta^2 x_1 y_1 + \theta(1-\theta)(x_1 y_2 + x_2 y_1) + (1-\theta)^2 x_2 y_2$$

$$\begin{cases} x_1 y_1 \geq 1 \\ x_2 y_2 \geq 1 \end{cases} \Rightarrow \begin{cases} \theta^2 x_1 y_1 \geq \theta^2 \\ (1-\theta)^2 x_2 y_2 \geq (1-\theta)^2 \\ \theta(1-\theta)(x_1 y_2 + x_2 y_1) \geq 2\theta(1-\theta) \end{cases}$$

$$\Rightarrow P'_x P'_y \geq \theta^2 + (1-\theta)^2 + 2\theta(1-\theta) = (\theta + 1 - \theta)^2 = 1$$

This set is convex



$$4. \{ (x, y) \in \mathbb{R}_{++}^2 \mid xy \leq 1 \}$$

find $P_1 = (x_1, y_1) = (1, 1)$
 ① $P_2 = (x_2, y_2) = (4, \frac{1}{4})$ both satisfy $xy \leq 1$

② take $\theta = \frac{1}{2}$, $P' = \theta P_1 + (1-\theta)P_2$

$$= [\theta x_1 + (1-\theta)x_2, \theta y_1 + (1-\theta)y_2]$$

$$= [\frac{1}{2} \times 1 + \frac{1}{2} \times 4, \frac{1}{2} \times 1 + \frac{1}{2} \times \frac{1}{4}]$$

$$= \cancel{(2.5, \frac{5}{8})} (\frac{5}{2}, \frac{5}{8})$$

$$P_x \cdot P_y = \frac{5}{2} \times \frac{5}{8} = \frac{25}{16} > 1$$

∴ This set is not convex

$$5. S = \{ x \in \mathbb{R}^n \mid x^T y \leq 1 \text{ for all } y \in C \}$$

① Pick x_1, x_2 from S

$$x_{\text{new}} = \theta x_1 + (1-\theta)x_2$$

② for $\forall y \in C$ $x_1^T y \leq 1$ and $x_2^T y \leq 1$

$$x_{\text{new}}^T y = \theta [x_1^T y] + (1-\theta)[x_2^T y] \leq \theta + 1-\theta = 1$$

∴ $x_{\text{new}}^T y \in S \Rightarrow S$ is convex

$$6. \{ x \in \mathbb{R}^n \mid x^T P^{-1} x \leq 1 \}$$

$$x_{\text{new}} = \theta x_1 + (1-\theta)x_2 = [\theta x_1 + (1-\theta)x_2]^T P^{-1} [\theta x_1 + (1-\theta)x_2]$$

$$= \theta^2 x_1^T P^{-1} x_1 + 2\theta(1-\theta)x_1^T P^{-1} x_2 + (1-\theta)^2 x_2^T P^{-1} x_2$$

$$\begin{cases} \theta^2 x_1^T P^{-1} x_1 \leq \theta^2 \\ (1-\theta)^2 x_2^T P^{-1} x_2 \leq (1-\theta)^2 \end{cases}$$

$$\Rightarrow x_{\text{new}} \leq \theta^2 + 2\theta(1-\theta) + (1-\theta)^2 = (\theta + 1-\theta)^2 = 1$$

$$x_1^T P^{-1} x_2 \leq \sqrt{(x_1^T P^{-1} x_1)(x_2^T P^{-1} x_2)} = 1$$

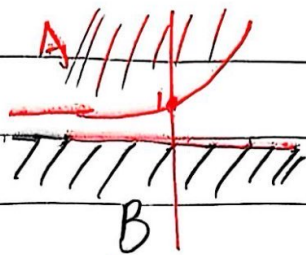
∴ This set is convex



Question 2

$$A = \{(x, y) \in \mathbb{R}^2 \mid y \geq e^x\}$$

$$B = \{(x, y) \in \mathbb{R}^2 \mid y \leq 0\}$$



① obviously A and B are ~~not~~ disjoint

② assumes exist $H = \{(x, y) \in \mathbb{R}^2 \mid ax + by = c\}$ strictly separated A and B

$$\begin{cases} ax + by > c & \forall (x, y) \in A \\ ax + by < c & \forall (x, y) \in B \end{cases}$$

(2.1) $H = ax + by = c$ can not intersect with B

$\Rightarrow H$ is parallel to B

$$H = ax = c \quad (c \neq 0) \quad (c > 0)$$

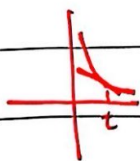
(2.2) let $H = ax = c = e^x \Rightarrow x = \ln c \quad (c > 0)$

exist $x = \ln c$ makes H intersect with B
which contradicts the definition of H

$\therefore$ This example satisfies requirements.

Q 3. 3.1 $\{(x, y) \in \mathbb{R}_+^2 \mid x_1 x_2 \geq 1\}$

for each tangent line at $(t, \frac{1}{t})$. $f(t) = \frac{1}{t}$, is convex



$$y - \frac{1}{t} \geq -\frac{1}{t^2}(x - t) \quad [1st-order-condition]$$

$$y \geq -\frac{1}{t^2}x + \frac{2}{t}$$

$$\bigcap_{t>0} \{(x, y) \in \mathbb{R}_+^2 \mid y \geq -\frac{1}{t^2}x + \frac{2}{t}\}$$



Q3

No.

Date.

$$3.2 \quad C = \{x \in \mathbb{R}^n \mid \|x\|_\infty \leq 1\}$$

$$\|x\|_\infty = \max_i |x_i| \leq 1$$

① $\hat{x}$ is the boundary of C

$$\Rightarrow \|\hat{x}\|_\infty = 1 \Rightarrow \exists i \in [1, \dots, n] \text{ such that } |\hat{x}_i| = 1$$

$$\therefore H = \{x \in \mathbb{R}^n \mid x_i = \hat{x}_i \text{ for each } i \in I\}$$

where $I = \{i \mid |\hat{x}_i| = 1\}$ is the set of indices where $\hat{x}$ attains maximum absolute value

$$3.3 \quad B = \{x \in \mathbb{R}^n \mid \|x\|_2 \leq 1\}$$

$$B = \bigcap_{\hat{x} \in S} \{x \in \mathbb{R}^n \mid \hat{x}^T x \leq 1\}$$

$$\text{where } S = \{x \in \mathbb{R}^n \mid \|x\|_2 = 1\}$$



No.

Q4-1

Date.

$$P(x, t) = \frac{x}{t}$$

$$P: \mathbb{R}^{n+1} \rightarrow \mathbb{R}^n \quad \text{dom } P = \{(x, t) \mid t > 0\}$$

where $x \in \mathbb{R}^n, t > 0$. Prove

1. $C \in \text{dom}(P)$ is convex $\Rightarrow P(C)$ is convex

~~pick~~ pick $y_1 = P_1(x_1, t_1), y_2 = P_2(x_2, t_2) = x_2/t_2$
 $= x_1/t_1$

~~To prove $P[\theta(x_1, t_1) + (1-\theta)(x_2, t_2)] = \theta y_1 + (1-\theta)y_2$~~

~~$P[\theta(x_1, t_1) + (1-\theta)(x_2, t_2)] =$~~

Prove $x = \theta(x_1, t_1) + (1-\theta)(x_2, t_2)$

$P(x) = \theta y_1 + (1-\theta)y_2$

① $P[\theta x_1 + (1-\theta)x_2, \theta t_1 + (1-\theta)t_2] = \frac{\theta x_1 + (1-\theta)x_2}{\theta t_1 + (1-\theta)t_2}$

$$= \frac{\theta x_1}{\theta t_1 + (1-\theta)t_2} + \frac{(1-\theta)x_2}{\theta t_1 + (1-\theta)t_2}$$

$$= \frac{\theta t_1}{\theta t_1 + (1-\theta)t_2} \cdot \frac{x_1}{t_1} + \frac{(1-\theta)t_2}{\theta t_1 + (1-\theta)t_2} \cdot \frac{x_2}{t_2}$$

$$= \lambda y_1 + (1-\lambda)y_2 \quad (\lambda = \frac{\theta t_1}{\theta t_1 + (1-\theta)t_2})$$

$\therefore P[\theta x_1 + (1-\theta)x_2, \theta t_1 + (1-\theta)t_2] = \lambda y_1 + (1-\lambda)y_2$

$\therefore P(C)$ is convex



Prove:

If $D \subset \mathbb{R}^n$ is convex, then $P^{-1}(D)$ is convex

let $\begin{cases} (x_1, t_1) \\ (x_2, t_2) \end{cases} \in P^{-1}(D)$, which is $\mathbb{R}^{n+1}$ vector.

$\begin{cases} u = \frac{x_1}{t_1} \\ v = \frac{x_2}{t_2} \end{cases} \in D$, which is $\mathbb{R}^n$ vector.

① D is convex, $\Rightarrow \lambda u + (1-\lambda)v \in D$

let $\lambda = \frac{\theta t_1}{\theta t_1 + (1-\theta)t_2}$ (θ is a constant).

$$\begin{aligned} \lambda u + (1-\lambda)v &= \frac{\theta t_1}{\theta t_1 + (1-\theta)t_2} \cdot \frac{x_1}{t_1} + \frac{(1-\theta)t_2}{\theta t_1 + (1-\theta)t_2} \cdot \frac{x_2}{t_2} \\ &= \frac{\theta x_1 + (1-\theta)x_2}{\theta t_1 + (1-\theta)t_2} \in D \end{aligned}$$

② Notice ~~$P^{-1}(\lambda u + (1-\lambda)v)$~~

$$P[\theta x_1 + (1-\theta)x_2, \theta t_1 + (1-\theta)t_2] = \lambda u + (1-\lambda)v$$

$$\Rightarrow [\theta x_1 + (1-\theta)x_2, \theta t_1 + (1-\theta)t_2] \in P^{-1}(D)$$

Overview, $\begin{cases} (x_1, t_1) \\ (x_2, t_2) \end{cases} \in P^{-1}(D) \Rightarrow [\theta x_1 + (1-\theta)x_2, \theta t_1 + (1-\theta)t_2] \in P^{-1}(D)$

$\therefore P^{-1}(D)$ is convex

